

# Omin Kwon

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## Research Interest

Computer Architectures & Machine Learning Systems, with an emphasis on co-designing ML algorithms and computer systems for efficient ML.

## Education

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| <p><b>M.S.</b>   <b>Seoul National University</b>, Computer Science and Engineering</p> <ul style="list-style-type: none"> <li>• Advisor: Prof. Jae W. Lee, ARC Lab</li> <li>• GPA: 4.3/4.3</li> </ul>                                                                                             | <p>Seoul, South Korea<br/>Sept 2025 – present</p> |
| <p><b>B.S.</b>   <b>Seoul National University</b>, Electrical and Computer Engineering (summa cum laude)</p> <ul style="list-style-type: none"> <li>• GPA: 4.26/4.3 (upper-division), 3.95/4.3 (overall)</li> <li>• Leave of absence for mandatory military service (May 2021-Feb 2023)</li> </ul> | <p>Seoul, South Korea<br/>Mar 2019 – Aug 2025</p> |
| <p><b>H.S.</b>   <b>Hansung Science High School</b>, Science</p>                                                                                                                                                                                                                                   | <p>Seoul, South Korea<br/>Mar 2017 – Mar 2019</p> |

## Publications

### SketchSSM: Write to the Full State, Read from a Compact Sketch

**Omin Kwon**, JoongWon Shin, Minseo Kim, Kurt Keutzer, Sehoon Kim, Jae W. Lee  
(Under review)

### HERALD: High Throughput Block Diffusion LLM Serving via CPU-GPU Cooperative KV Cache Retrieval

**Omin Kwon**, Doyeon Kim, Jongseok Park, Seung Yul Lee, Ion Stoica, Jae W. Lee  
[arxiv.org/abs/2606.21633](https://arxiv.org/abs/2606.21633) (Under review)

### ELIXIR: Disaggregating FFN, Attention, and KV Cache via CXL-PNM for Elastic LLM Serving

**Omin Kwon\***, KyungSoo Kim\*, Haneul Park, JoongWon Shin, Nachuan Wang, UGyeong Song, Yeonhong Park, Nam Sung Kim, Jae W. Lee  
(Under review)

### MAGE: All-[MASK] Block Already Knows Where to Look in Diffusion LLM

**Omin Kwon**, Yeonjae Kim, Doyeon Kim, Minseo Kim, Yeonhong Park, Jae W. Lee  
[arxiv.org/abs/2602.14209](https://arxiv.org/abs/2602.14209) (NeurIPS 2026; ICML 2026 AdaptFM Workshop (Best Paper Award))

### STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer

**Omin Kwon\***, Kyungjun Oh\*, Jaeyong Lee, Myungsuk Kim, Jihong Kim  
[doi.org/10.1109/ICCAD66269.2025.11240772](https://doi.org/10.1109/ICCAD66269.2025.11240772) (ICCAD 2025)

### NestedFP: High-Performance, Memory-Efficient Dual-Precision Floating Point Support for LLMs

Haeun Lee, **Omin Kwon**, Yeonhong Park, Jae W. Lee  
[neurips.cc/virtual/2025/loc/san-diego/poster/117613](https://neurips.cc/virtual/2025/loc/san-diego/poster/117613) (NeurIPS 2025)

## AiDE: Attention-FFN Disaggregated Execution for Cost-Effective LLM Decoding on CXL-PNM

KyungSoo Kim, **Omin Kwon**, Yeonhong Park, Jae W. Lee

[doi.org/10.1109/LCA.2025.3597323](https://doi.org/10.1109/LCA.2025.3597323) (IEEE Computer Architecture Letters (CAL))

## Experiences

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| <b>Seoul National University</b> , Teaching Assistant, Advanced Computer Architecture (4190.571)                                                                   | Seoul, South Korea<br>Sept 2026 – present  |
| <b>University of Illinois Urbana-Champaign (UIUC)</b> , Visiting Researcher <ul style="list-style-type: none"><li>FAST Lab, hosted by Prof. Nam Sung Kim</li></ul> | Urbana, IL, USA<br>Mar 2026 – May 2026     |
| <b>Seoul National University</b> , Microelectronics (430.207B) Tutor                                                                                               | Seoul, South Korea<br>Mar 2025 – June 2025 |
| <b>Seoul National University</b> , Digital System Design (430.315) Tutor                                                                                           | Seoul, South Korea<br>Sept 2024 – Dec 2024 |
| <b>Seoul National University</b> , Computer Organization and Design (430.322) Tutor                                                                                | Seoul, South Korea<br>Mar 2024 – June 2024 |
| <b>Republic of Korea Air Force</b> , Military Service                                                                                                              | South Korea<br>May 2021 – Feb 2023         |

## Presentations

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| <b>MAGE: All-[MASK] Block Already Knows Where to Look in Diffusion LLM</b><br>Oral presentation at the ICML 2026 AdaptFM Workshop. <a href="#">arXiv</a> .                    | 2026 |
| <b>STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer</b><br>Oral presentation at ICCAD 2025, Munich. <a href="#">Paper</a> . | 2025 |

## Honors & Awards

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| <b>Best Paper Award @ ICML26 AdaptFM Workshop</b><br>1 Best Paper among 123 accepted papers from 218 submissions.                                                                                                                                                      | July 2026           |
| <b>Grand Prize (Minister of Science and ICT Award), 2024 AI Semiconductor Idea and Design Contest</b><br>Designed a resolution-aware diffusion accelerator leveraging computational staleness.<br>News: <a href="#">Edaily</a> .                                       | Jan 2025            |
| <b>Grand Prize (Deputy Prime Minister and Minister of Education Award), 2024 STOB League Semiconductor Contest</b><br>Designed a state-aware data randomizer for high-performance SSDs, improving performance and device lifetime. News: <a href="#">Dong-A Ilbo</a> . | Dec 2024            |
| <b>Grand Prize, 2024 FriendliAI LLM Hackathon</b><br>Developed a knowledge-graph RAG application using the FriendliAI API.                                                                                                                                             | July 2024           |
| <b>SNU Specialized Semiconductor College Scholarship</b>                                                                                                                                                                                                               | Aug 2024 – Aug 2025 |
| <b>Merit-Based Scholarship</b>                                                                                                                                                                                                                                         | 2023, 2024          |